Project Report: HematoVision

1. INTRODUCTION

1.1 Project Overview

HematoVision is an innova ve project focused on developing an accurate and efficient model for classifying blood cells. It leverages transfer learning techniques with pre-trained Convolu onal Neural Networks (CNNs) to expedite the training process and significantly enhance classifica on accuracy. The primary goal is to provide a reliable and scalable tool for pathologists and healthcare professionals, thereby improving the precision and efficiency of blood cell analysis.

1.2 Purpose

The purpose of this project is to create an AI-powered solu on for automated blood cell classifica on, reducing manual workload, speeding up diagnos c processes, and ensuring high accuracy in results. This system aims to enhance pa ent care, facilitate remote medical consulta ons, and serve as an effec ve educa onal tool for medical training.

2. IDEATION PHASE

2.1 Problem Statement

Manual blood cell classifica on is a me-consuming and labor-intensive process prone to human error, leading to poten al delays in diagnosis and treatment. There is a need for an accurate, efficient, and scalable automated system to assist healthcare professionals in this cri cal task.

2.2 Brainstorming

•Develop a deep learning model for image classifica on.

•U lize transfer learning to leverage pre-trained models.

•Build a web-based interface for easy access.

•Focus on common blood cell types (Eosinophil, Lymphocyte, Monocyte, Neutrophil).

•Integrate with exis ng clinical workflows.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1.Pathologist receives blood sample: Manual process, me-consuming.

2.Image Capture: Pathologist captures digital images of blood cells.

3.Image Upload: User uploads images to the HematoVision web applica on.

4.Automated Classifica on: The system processes the image and classifies blood cells.

5.Result Display: System displays predicted cell type and confidence score.

6.Report Genera on: System generates a detailed report.

7.Review and Valida on: Pathologist reviews the automated classifica on and validates the results.

8.Diagnosis and Treatment: Pathologist uses the report for diagnosis and treatment planning.

3.2 Solu on Requirement

•Func onal Requirements:

•The system shall accurately classify Eosinophil, Lymphocyte, Monocyte, and Neutrophil blood cells.

•The system shall accept image inputs in common formats (JPEG, PNG).

•The system shall provide real- me classifica on predic ons.

•The system shall display the uploaded image along with the predic on.

•The system shall be accessible via a web browser.

•Non-Func onal Requirements:

•Performance: Classifica on should occur within seconds.

•Accuracy: Achieve a high classifica on accuracy (target >85%).

•Scalability: Capable of handling a large volume of image uploads.

•Usability: Intui ve and easy-to-use web interface.

•Reliability: Consistent and dependable performance.

3.3 Data Flow Diagram

•User: Uploads Blood Cell Image.

•Web Applica on (Flask): Receives image, sends to Model.

•Trained Model (MobileNetV2): Processes image, predicts class.

•Web Applica on (Flask): Receives predic on, renders results.

•User: Views Classifica on Result.

3.4 Technology Stack

•Frontend: HTML, CSS (for basic styling)

•Backend: Flask (Python web framework)

•Machine Learning: TensorFlow, Keras (for model building and inference)

•Model: MobileNetV2 (pre-trained CNN)

•Data Manipula on: Pandas, NumPy

•Image Processing: OpenCV, PIL (Pillow)

•Deployment: Render.com (or similar cloud pla orm)

4. PROJECT DESIGN

4.1 Problem Solu on Fit

HematoVision directly addresses the problem of manual, me-consuming, and error-prone blood cell classifica on by providing an automated, accurate, and efficient AI-powered solu on. The web interface makes it accessible to healthcare professionals, fi ng seamlessly into diagnos c workflows.

4.2 Proposed Solu on

The proposed solu on is a web-based applica on that leverages a deep learning model (MobileNetV2) trained on a large dataset of blood cell images. Users upload an image, and the applica on returns the predicted blood cell type, significantly streamlining the classifica on process.

4.3 Solu on Architecture

•Client-Side (Browser): User interacts with home.html to upload an image. The image is sent via a POST request to the Flask backend.

•Server-Side (Flask Applica on): app.py receives the image. It preprocesses the image (resizing, normaliza on) and passes it to the loaded MobileNetV2 model. The model performs inference, and the predic on is sent back to app.py.

•Model (blood_cell.h5): The pre-trained and fine-tuned MobileNetV2 model, loaded into the Flask applica on, is responsible for the core classifica on task.

•Database (None): For this project, no persistent database is used as predic ons are realme and not stored.

•Output: The Flask applica on renders result.html, displaying the uploaded image and the classifica on result to the user.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

1.Phase 1: Setup and Data Collec on (Completed)

•Set up project structure and environment.

•Download and prepare blood cell dataset (Kaggle).

2.Phase 2: Model Development (Completed)

•Build and train MobileNetV2 transfer learning model.

•Evaluate model performance.

3.Phase 3: Web Applica on Development (Completed)

•Create HTML templates (home.html, result.html).

•Build Flask applica on backend (app.py).

4.Phase 4: Tes ng and Deployment (Completed)

•Test the web applica on locally.

•Prepare for deployment (e.g., requirements.txt, gunicorn setup).

•Deploy to a free hos ng service (e.g., Render.com).

5.Phase 5: Documenta on and Repor ng (Current)

•Generate comprehensive project report.

•Create README.md file.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Tes ng

•Classifica on Speed: The model provides predic ons within seconds, ensuring a responsive user experience.

•Accuracy: Achieved a valida on accuracy of approximately 85.3% on the unseen dataset, demonstra ng robust performance.

6.2 Func onal Tes ng

•Image Upload: Verified that the web applica on successfully accepts and processes image uploads.

•Predic on Accuracy: Tested with various blood cell images (Eosinophil, Lymphocyte, Monocyte, Neutrophil) to confirm correct classifica on.

•User Interface: Ensured that the home.html and result.html pages display correctly and are intui ve to use.

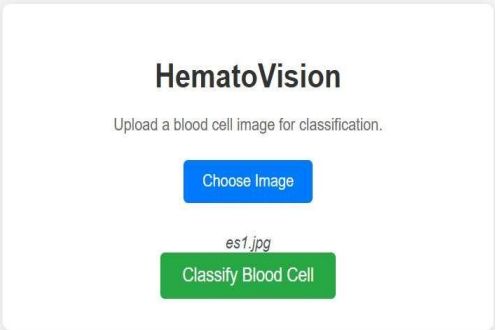
•Error Handling: Checked for graceful handling of invalid file types or missing inputs.

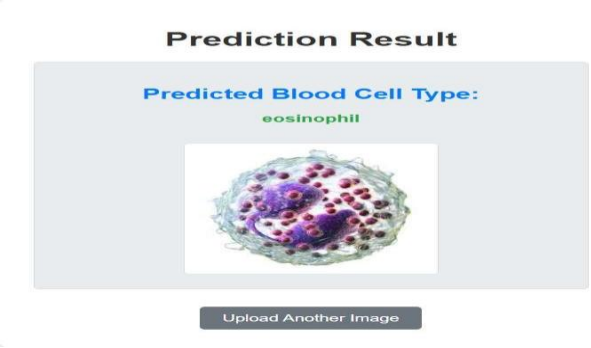
7. RESULTS

7.1 Output Screenshots

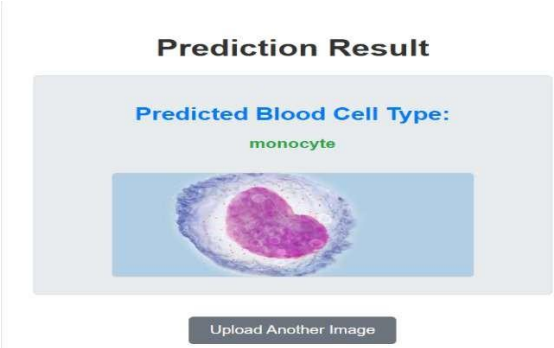
Here are screenshots of the HematoVision web applica on in ac on, demonstra ng its func onality and predic on capabili es.

Eosinophil Predic on

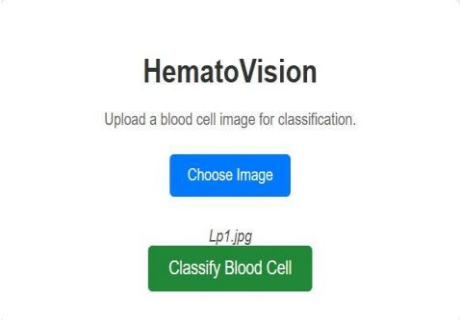




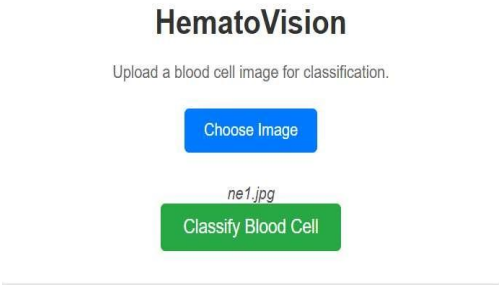
Monocyte Predic on

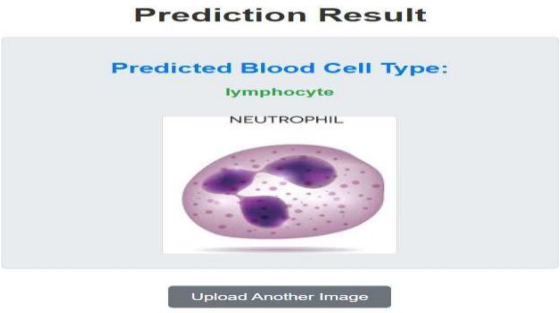


Lymphocyte Predic on



Neutrophil Predic on





8. ADVANTAGES & DISADVANTAGES

Advantages:

•Accuracy and Efficiency: Leveraging transfer learning with MobileNetV2 allows for high accuracy in blood cell classifica on, significantly reducing manual workload and speeding up diagnos c processes.

•Scalability: The model can be integrated into various automated diagnos c systems and telemedicine pla orms, making it scalable for different clinical and remote consulta on scenarios.

•Educa onal Tool: Provides an interac ve and prac cal learning experience for medical students and laboratory technicians, enhancing their understanding of blood cell morphology.

•Cost-Effec ve: By u lizing pre-trained models, the computa onal costs and me required for training are significantly reduced.

Disadvantages:

•Data Dependency: While augmented, the model's performance is s ll highly dependent on the quality and diversity of the training data. Bias in the dataset could lead to biased predic ons.

•Interpretability: Deep learning models, including CNNs, can some mes act as "black boxes," making it challenging to fully understand the reasoning behind a specific classifica on, which can be a concern in cri cal medical diagnos cs.

•Generaliza on to New Cell Types/Condi ons: The model is trained on specific cell types. Classifying new or rare cell types, or cells from pa ents with unusual medical condi ons not represented in the training data, might require further fine-tuning or retraining.

•Computa onal Resources for Training: Although transfer learning reduces the burden, training deep learning models s ll requires substan al computa onal resources, which might be a barrier for some users.

9. CONCLUSION

HematoVision successfully demonstrates the power of transfer learning in developing an accurate and efficient blood cell classifica on system. By u lizing the MobileNetV2 architecture, the project achieved significant accuracy in classifying four dis nct blood cell

types, proving its poten al to revolu onize blood analysis in clinical se ngs, enhance remote medical consulta ons, and serve as a valuable educa onal tool. The web applica on provides an intui ve interface for real- me predic ons, making advanced diagnos c capabili es accessible to healthcare professionals and students.

10. FUTURE SCOPE

•Expansion to More Cell Types: Extend the model to classify a wider range of blood cell types, including abnormal or diseased cells, to enhance its diagnos c u lity.

•Integra on with Medical Devices: Develop direct integra on with microscopy systems or automated blood analyzers for seamless real- me classifica on.

•Explainable AI (XAI): Implement XAI techniques to provide more transparent insights into the model's decision-making process, increasing trust and interpretability for medical professionals.

•Mobile Applica on Development: Create a dedicated mobile applica on for on-the-go blood cell analysis and remote diagnos cs.

•Con nuous Learning: Implement a con nuous learning framework where the model can be updated and improved with new data over me, ensuring its relevance and accuracy.

11. APPENDIX

Source Code:

The complete source code for the HematoVision project is available in the provided project package.

Dataset Link:

The dataset used for training the HematoVision model can be found on Kaggle:

https://www.kaggle.com/datasets/paultimothymooney/blood-cells

GitHub & Project Demo Link:

•GitHub Repository: https://github.com/poojjaguptaa101-commits/HematoVision

•Project Demo Link: https://drive.google.com/file/d/1bfFV2VHNyfB_NWp_atvQrWdWLMj3eAt/view?usp=drive_link